

Backward Reasoning in Isabelle (BW)

Lecture Notes

Course: Computer-Assisted Theorem Proving (7CCSACTP)

1. Context & Comparison with Forward Reasoning

→ **Last Video (Forward Reasoning):**

- Given a rule/fact with premises ($P \implies Q$), instantiate those premises using known facts to derive new results forward (Q).

→ **This Video (Backward Reasoning):**

- Reasoning from the target goal backwards to required premises using rules (`apply (rule ...)`).

2. Intuition & Mechanics of Backward Reasoning

Core Intuition:

- Suppose your target goal is P .
- If you have a lemma `lem1`: $Q \vdash P$ (or $Q \implies P$), backward reasoning replaces the goal P with the simpler/new subgoal Q .
- **In Isabelle:** Executed via the `rule` proof method.

Example Execution:

Rule (lem1):	$Q \vdash P$
Current Goal:	<code>goal : P</code>
Proof Method:	<code>apply (rule lem1)</code>
New Subgoal:	<code>goal : Q</code> ✓

3. Connectives, Advantages & Next Topics

- **Operator Introduction Rules:** If the goal contains a logical connective "*op*" (e.g. $\wedge, \vee, \rightarrow, \forall$), analyze/decompose that goal using the corresponding *op-introduction* rule.
- **Why Use Backward Reasoning?**
 - **Debugging:** Allows clear visibility and debugging of proof goals.
 - **Top-Down Development:** Enables structured, top-down proof construction.
- **Next Video Topic:** Proofs by substitution (`subst`).